Technical Datasheet Input / Output Modules with Modbus TCP Protocol with Ethernet Interface

The IO module communicates via Ethernet Interface. The module is designed to allow interface with other Modbus TCP/IP compatible devices. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.
Digital IO module is sturdy, low power usage and easy to use.

8 Port AI Module with Modbus TCP Ethernet Port: -

The IO modules are mounted on DIN rail mountable casing.

The design of the modules incorporates ‘**resettable Fuses**’ to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 70 mm B x 50 mm H
Power	Input Power – 12 – 24 VDC or 24 V AC/DC Typical – 12V DC @ 80mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	

AI Inputs –

Channels	8
Input Signal	4 – 20 mA / 0 – 20 mA / 0 – 10 V (jumper selectable)
Accuracy	± 2% Full scale
Input Resolution	10-bit /12-bit resolution (optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Additional Features: -

Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
Web based Configuration needed for the IO board

Configuration Settings: -

Communication Speed	10/100 Base-T Ethernet compatible Interface
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	(Web Selectable)
Function Code AI	0x03 Read Holding Registers
AI Register Address	**10 Bit** -1,2,3,4,5,6,7,8 / **12 Bit** – 9,10,11,12,13,14,15,16

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	**AI 1 – 10 Bit**	40001	0X03	RS485	16 Bit Unsigned int
2	**AI 2 – 10 Bit**	40002	0X03	RS485	16 Bit Unsigned int
3	**AI 3 – 10 Bit**	40003	0X03	RS485	16 Bit Unsigned int
4	**AI 4 – 10 Bit**	40004	0X03	RS485	16 Bit Unsigned int
5	**AI 5 – 10 Bit**	40005	0X03	RS485	16 Bit Unsigned int
6	**AI 6 – 10 Bit**	40006	0X03	RS485	16 Bit Unsigned int
7	**AI 7 – 10 Bit**	40007	0X03	RS485	16 Bit Unsigned int
8	**AI 8 – 10 Bit**	40008	0X03	RS485	16 Bit Unsigned int
9	**AI – 1 12 Bit**	40009	0x03	RS485	16 Bit Unsigned int
10	**AI – 2 12 Bit**	40010	0x03	RS485	16 Bit Unsigned int
11	**AI – 3 12 Bit**	40011	0x03	RS485	16 Bit Unsigned int
12	**AI – 4 12 Bit**	40012	0x03	RS485	16 Bit Unsigned int
13	**AI – 5 12 Bit**	40013	0x03	RS485	16 Bit Unsigned int
14	**AI – 6 12 Bit**	40014	0x03	RS485	16 Bit Unsigned int
15	**AI – 7 12 Bit**	40015	0x03	RS485	16 Bit Unsigned int
16	**AI – 8 12 Bit**	40016	0x03	RS485	16 Bit Unsigned int

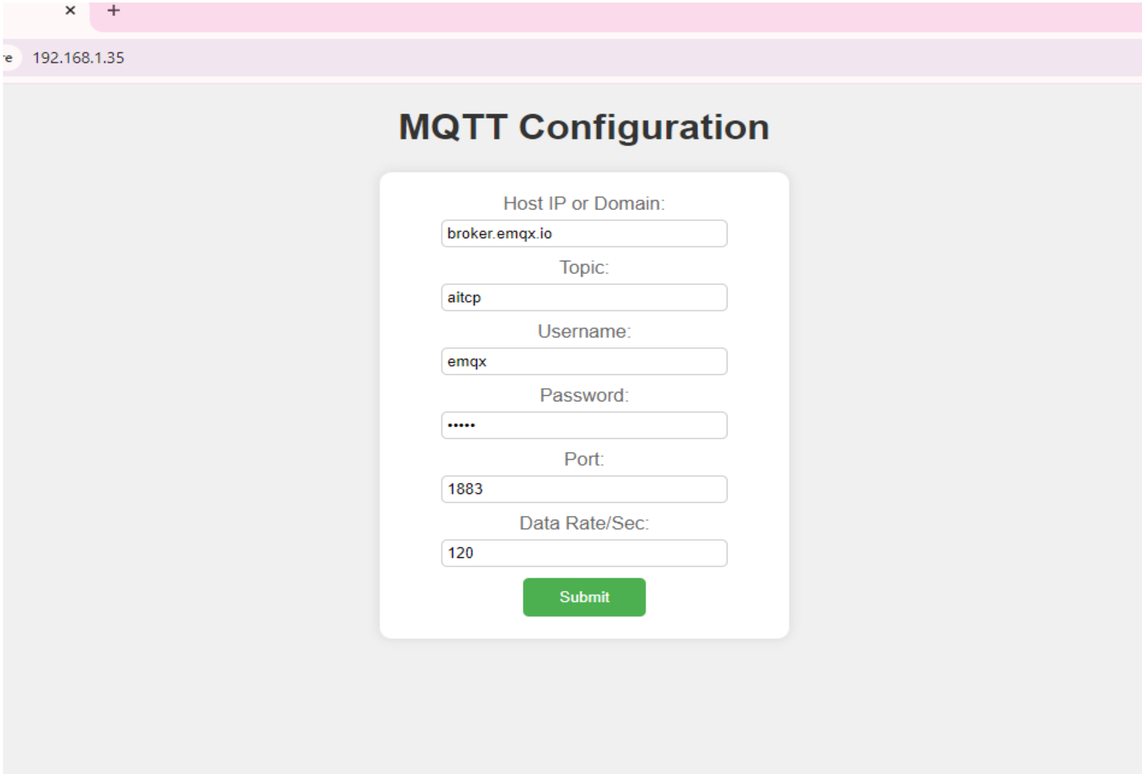
Note: -

For MODBUS communications, **a shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

- **Enter the IP address of the server in your web browser, this type of page will open.**
- **ip address for the server is 192.168.1.35**



Specification.

1. **Hosting Capacity:**

- 8 Analog Pins.

2. **Interface:**

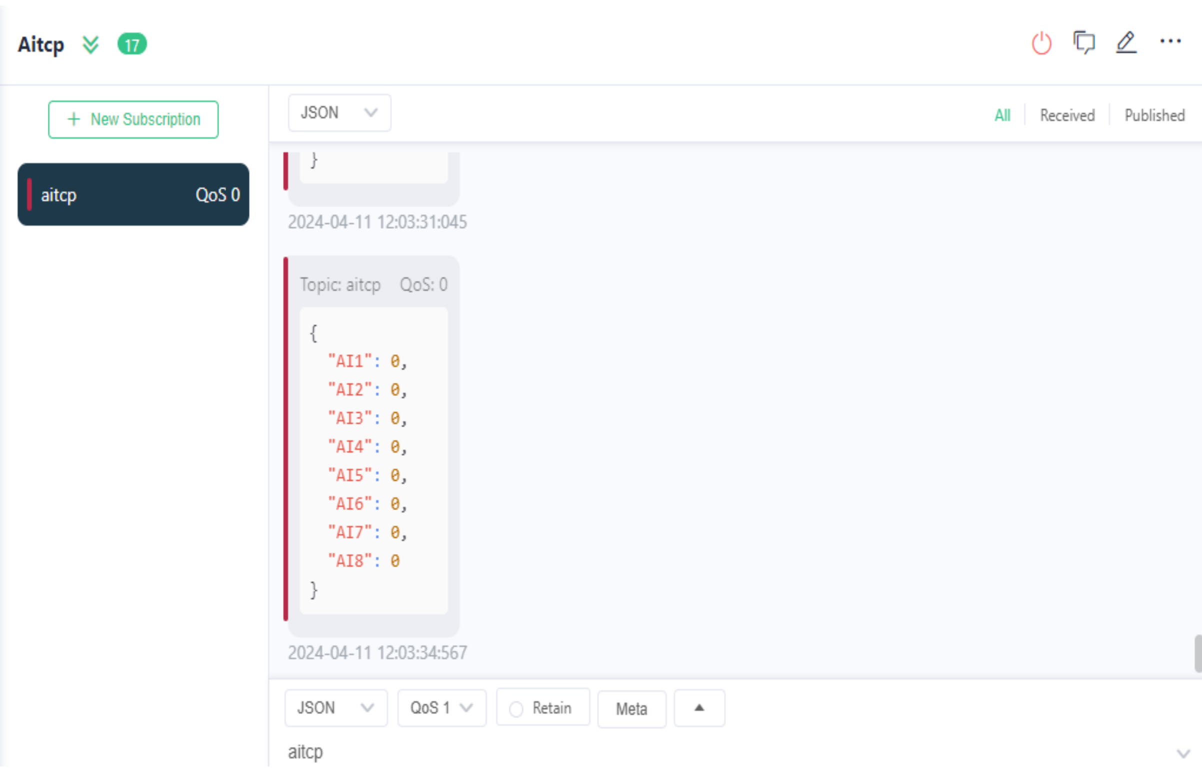
- A user Friendly Interface for users.

  1. Host Ip or Domain
  2. Topic.
  3. Username.
  4. Password
  5. Data Rate/Sec

Instruction:

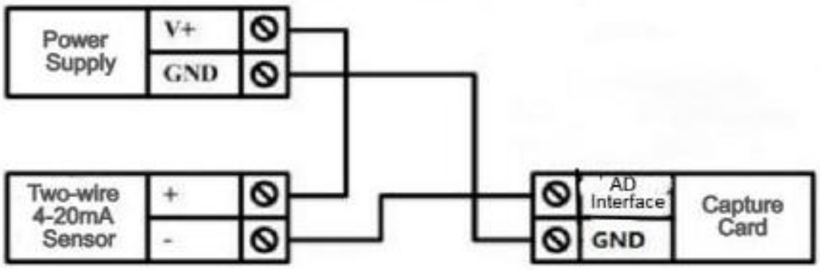
Host Ip or Domain: Input the address of your MQTT broker.
Topic: Define the MQTT topic to which your device will subscribe or publish messages.
Username: Provide the username associated with your MQTT broker
Password: Enter the password corresponding to your MQTT broker account.
Port: Specify the port number for the MQTT connection (default is usually 1883).
Dara Rate/Sec: Define the data rate at which data will be transmitted or updated. It is important to input the data rate in seconds when entering it.

After Entering the details click on the Submit Button
Open the MQTT App or Browser You will get data in the given format as per showing in the image.

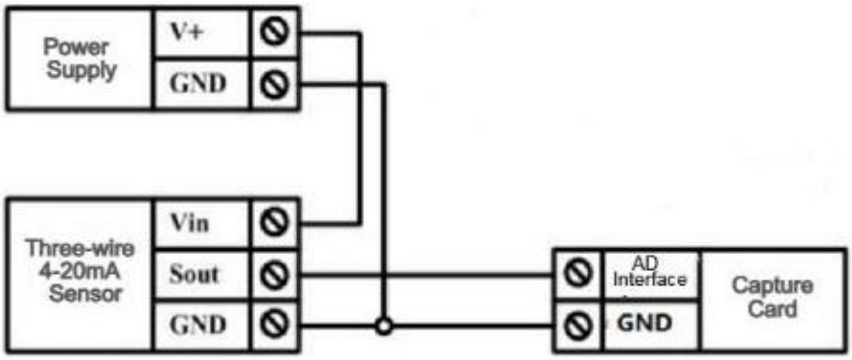


Analog pin AI1 to AI8 are displayed and data corresponding to each pin will be shown in front of them.

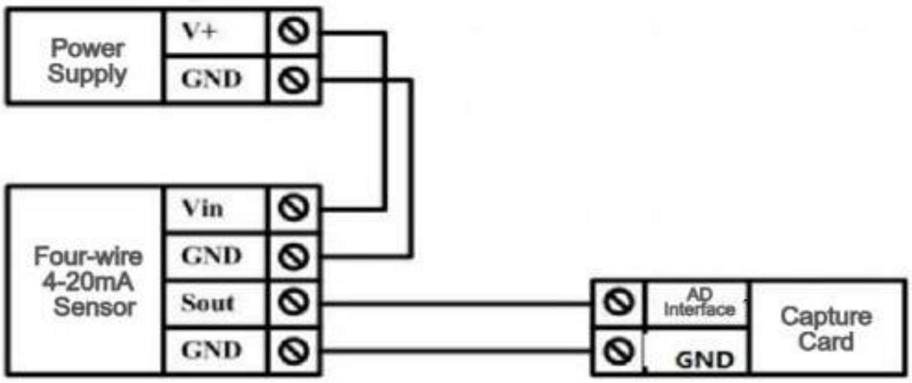
Two-wire Sensor Wiring Diagram



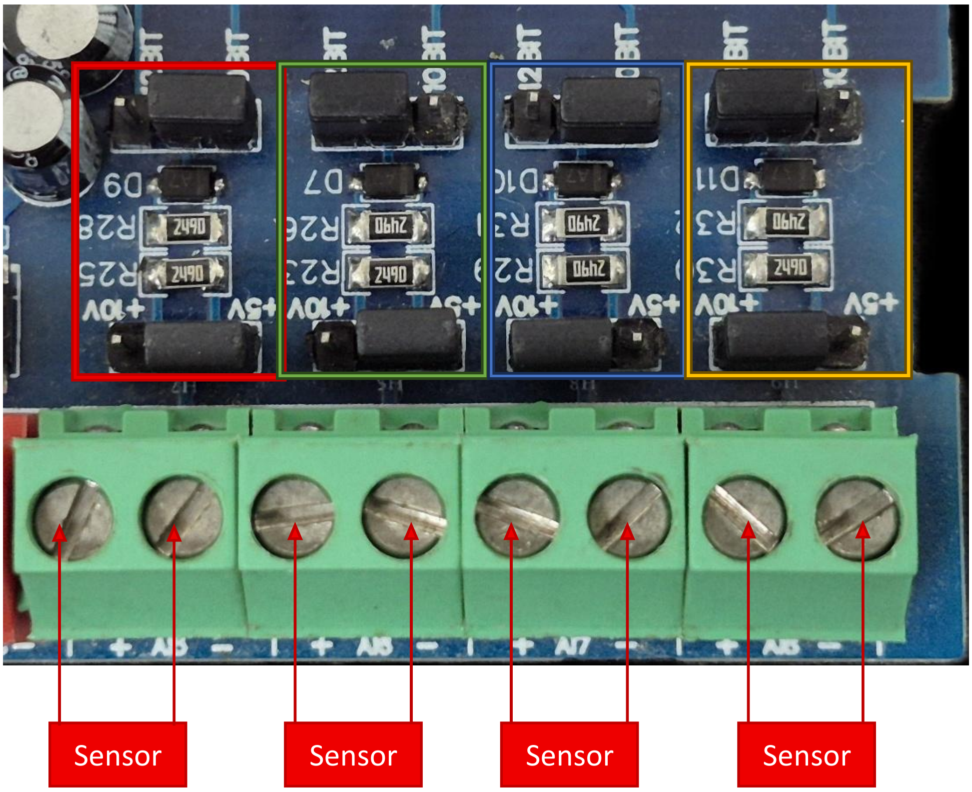
Three-wire Sensor Wiring Diagram



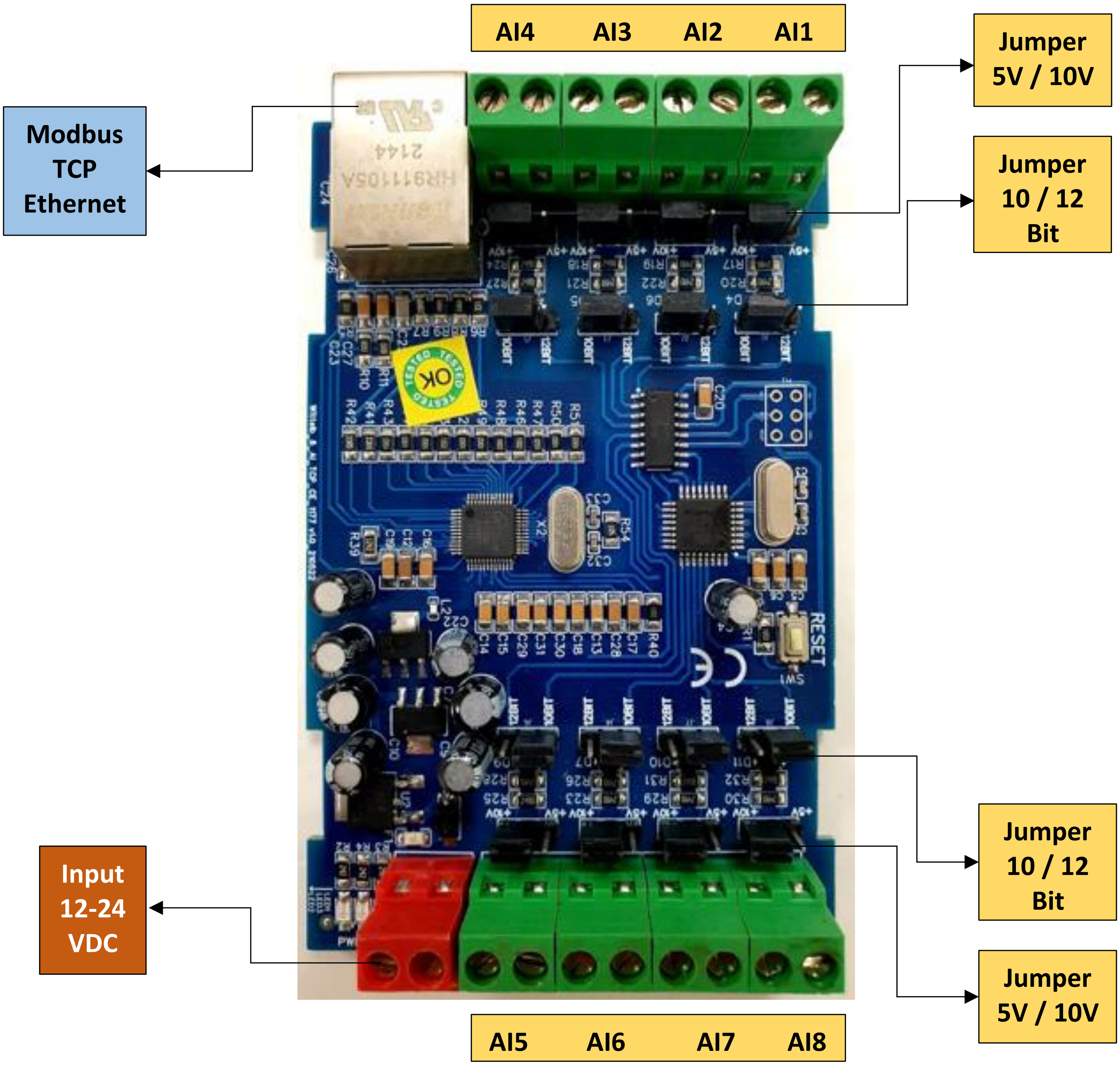
Four-wire 4-20mA Sensor



AI



- **Analog Input (AI) Operation:** Each channel provides two terminals: **+** (Positive) and **-** (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
    - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
    - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
    - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
    - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
    - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
    - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.



NOTE – By default the Jumper setting on the board is 10 V and 10 Bit.

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